

Towards Energy-aware Communications: Connecting the power grid and communication infrastructure

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ABSTRACT

This paper introduces the motivation and main ideas from the Flexcomm project, which aims to conceive and develop strategies for energy-aware communications in Core & Metro networks. The innovative aspect of this project consists of integrating communication systems with energy distribution systems as an enabler to network protocols able to adjust data routing according to the energy flexibility and source.

1 THE BIG PICTURE

The concept of Sustainable Smart Cities (SSC) is based on innovative and pervasive uses of Information and Communication Technologies (ICT) towards improved population quality of life and efficient urban operations. The Internet of Things (IoT) is at the core of any smart city design, including sensing and actuators devices along with multilayer network devices in a complex system supporting advances in various services and infrastructures. Globally, by 2025 it will encompass between 50 to 100 billion devices, representing a market reaching US\$ 1.7 trillion with a significant portion allocated to network infrastructures responsible for transporting around 24EB monthly [2, 4].

Despite being used as a platform supporting energy efficiency in diverse scenarios, these ultra-dense and heterogeneous networks pose serious energy consumption concerns by themselves. Recent estimations show that the ICT sector accounts for 5% to 9% of global electricity consumption and more than 2% of greenhouse gas emissions¹. Half of this amount is consumed by data transmission networks, which is expected to increase up to 310TWh by 2031².

Considering the distributed nature of the access network in next-generation cellular technology, i.e., 5G, and the current scenario of mobile data usage, which grows by a factor of 10 every 5 years¹, imposes challenges up to 10 fold compared with 4G to fulfil the energy reduction milestone. Therefore, the way such a huge amount of data is transmitted assumes a central role in managing natural resources.

Typically, the strategies used by network operators toward energy-efficient communications fall into three groups: (i) replacing existing network assets with more efficient versions; (ii) turning off spare devices whose capacity is not required to transport off-peak traffic; and (iii) optimising traffic engineering processes. Although effective, strategy (i) faces economic challenges, as replacing core devices can increase operational costs to unacceptable levels. Strategy (ii) has demonstrated promising results, e.g., up to 23% of energy reduction in backbone networks [1]. However, it tends to be less effective for access networks due to last-mile connections' distributed and dynamic nature. Strategy (iii) is commonly used along with strategy (ii) as a mechanism to maintain acceptable Quality-of-Service (QoS) levels when devices are turned off, and traffic needs to be transported via alternative routes.

The growing deployment of Software-Defined Networks (SDN) has fostered research toward efficient traffic engineering. These strategies are essential for reducing or slowing the energy consumption growth in Wide Area Networks (WAN). However, it does not mean they are reducing the underlying carbon footprint. Typically, deciding which devices will be turned off and which alternative route will be used is related to operational costs and QoS, not the energy source or distribution constraints.

At the same time, the energy market, sponsored by Distribution System Operators (DSO), is currently undergoing a transformation, accounting for the increasing volume of Distributed Renewable Energy Sources (DRES) and the digitalisation of all processes. Essential for DRES is the concept of *Flexibility*, identifying the zones or nodes of the grid that require activation or deactivation to ensure QoS. Moreover, digital twinning, e.g., virtual power plants [3], and the provision of new energy and non-energy data-driven services are quickly changing how the grid is monitored and controlled, where flexibility provides a conceptually good candidate for interfacing with systems that impose high consumption. This close relation enables a promising interconnection that is just beginning to be explored and is at the core of Flexcomm.

¹ Source: Ericsson, 2020 - "Breaking the curve: An innovative approach to reducing mobile network energy use"

²Source: IEA-4E EDNA, 2019 - "Intelligent efficiency for data centres & wide area networks"

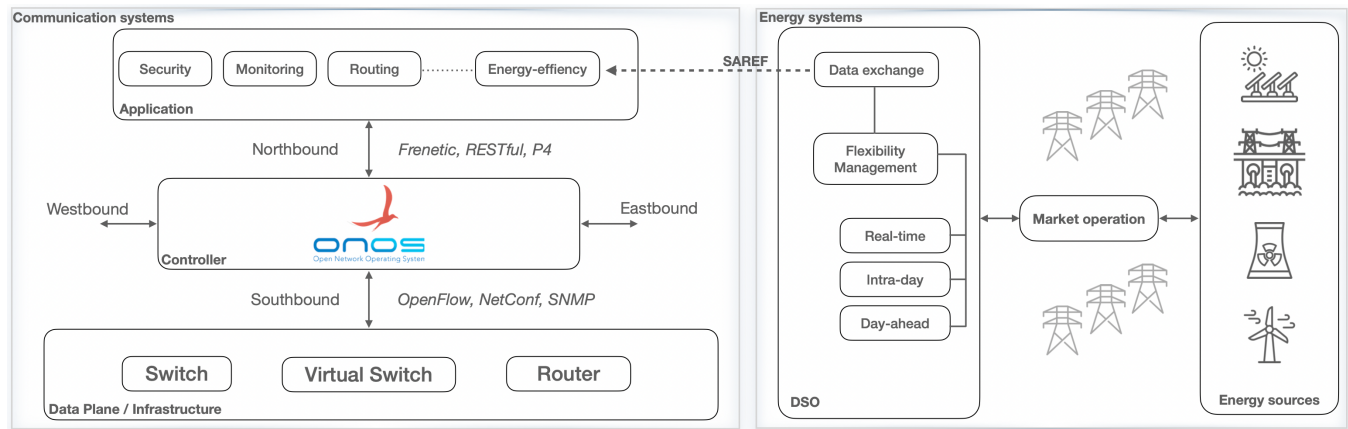


Figure 1: Architectural scheme of infrastructures' integration

2 VISION AND TASKS

The main objective pursued in Flexcomm is to endow current and forthcoming communication systems to adapt dynamically and autonomously to energy conditions and sources. Notice that energy-aware communications here stated exceed the concept of lightweight communications. The latter operates towards energy conservation, regardless of the power source.

The work's vision is based on the idea that routing algorithms can be enhanced to consider energy conditions and sources within their routing processes. Here, the energy conditions relate to the real-time power consumption of network devices and the energy availability announced by DSO through flexibility-related data exchange API. As briefly illustrated in Figure 1, DSO can share with higher consumption clients their geographically distributed energy availability for the next day and the actual real-time condition. ISP can use this information to adjust SDN-based routing according to the traffic demand. Ideally, traffic will be routed preferably through topology sections with higher energy availability or powered by renewable sources³.

As research in production environments is challenging and costly, Flexcomm is developing a full-fledged simulation solution to support flexibility in solution design and Proofs-of-Concept (PoC). A summary of the project outcomes includes:

Energy-aware simulator modules that extend the *ns-3* simulator by including a generic energy model for network devices, an OpenFlow interface, and a real-time report module;

Convergent information model component for SDN controllers responsible for mapping data retrieved from

the power grid into a local representation based on the IPFIX information model;

Convergent SDN controller extending the *ONOS* controller to *i*) communicate with data plane in network infrastructure and public API in power systems; *ii*) support routing algorithms able to combine this data in traffic engineering and selective device shutdown;

Energy-aware OSPF PoC SDN implementation of an original OSPF routing protocol designed to consider the power grid condition and source when calculating route costs;

Module for automated test component responsible for mapping an XML configuration file in test environments, including network topology, routing devices, energy model, traffic load, and power supply conditions.

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³Currently, DSO only shares the next-day availability in blocks of one hour without indicating the energy source.